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AI IN HEALTHCARE
SYMPOSIUM

Reducing Cognitive Load in Clinical Decision-Making with GenAI-Powered Report Analysis

— a prototype and study proposal

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INTRODUCTION

Clinicians managing complex patients must manually navigate multiple longitudinal test reports — a cognitively demanding process that makes it difficult to identify trends and synthesise findings at the point of care. The burden of integrating these findings can lead to missed insights and suboptimal clinical decisions.

AIMS

To develop a system that automatically extracts, summarises, and visualises trends from longitudinal patient test reports (structured and unstructured data), and surfaces personalised best-practice clinical recommendations. Evaluated here on respiratory function tests and echocardiograms.

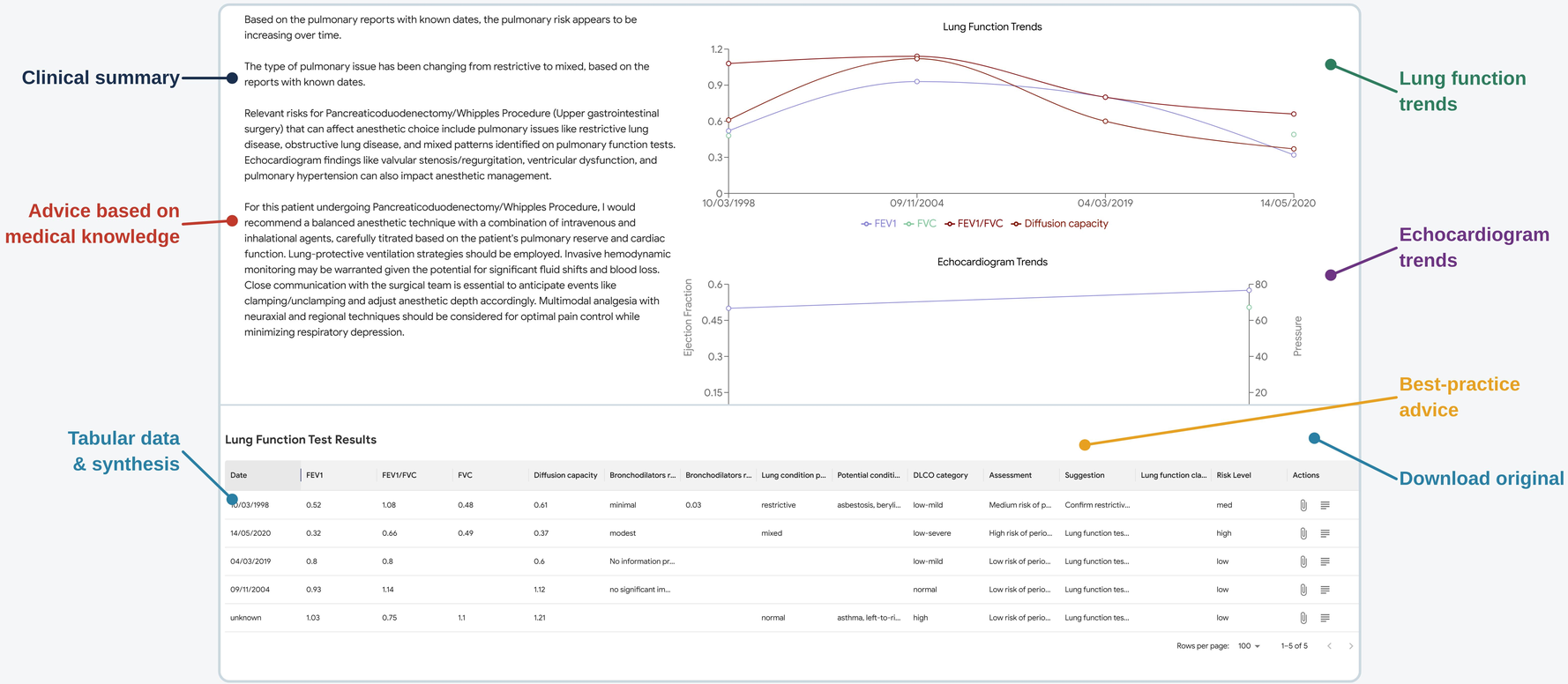
METHODS

A model-agnostic LLM iteratively processes patients' test reports to extract data points for key clinical metrics, plot trends and create a summary. Prompts include guardrails to avoid hallucinations. Best-practice recommendations are codified and dynamically presented via a personalised rule-based look-up.

SYSTEM OUTPUTS (PER REPORT MODALITY)

- Tabular extraction of numerical *and* non-numerical trends
- Synthesis: Assessment · Classification · Risk Level · Suggestions · Actions

Reports per patient → **LLM** Analyser → **Plots & Summaries** → **Clinical Review**



RESULTS

The system analysed reports, extracted key information, presented trends over time in plot and table formats, provided text-based summaries and gave best-practice clinical recommendations. Results were reviewed by an experienced anaesthetist and found to be accurate.

CONCLUSIONS

This proof-of-concept demonstrates that a GenAI-based system can accurately synthesise longitudinal patient reports, visualise results, and surface personalised best-practice recommendations, significantly reducing clinician cognitive load. A prospective clinical trial is warranted.

FUTURE WORK

- RAG-based dynamic best-practice guideline lookup
- Clinician chat interface for interactive report discussion
- Prospective clinical trial evaluation

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